DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-200-03 — Replace VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1021/A2)

Area	A200 Vessels and column	**Task**	VR Valve replacement in kind
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	**V-102** — V-102 vertical vessel 72" ID × 216" T/T — E 53'-6" N -25'-0" EL 100'-8-1/4"		
Equipment	V-102	**Lines**	4"-P300-P-1021
Isometrics	ISO-4"-P300-P-1021 (generated); owner's drawings ISO-P-1021		
Permits	General work permit		

Reason

Outlet isolation valve stem packing leaking and seat passing.

Scope of work

1. Isolate per the isolation list, drain and vent 4"-P300-P-1021 between the isolations.
2. Unbolt and remove VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1021/A2) at E 53'-6" N -20'-8-3/16" EL 109'-0".
3. Inspect both flange faces (SW-NDE-5); dress or report damage.
4. Install a new VALVE, GATE, WEDGE, OS&Y 4" P300 of the same size, class and face-to-face (GATE4-P300), with new gaskets and studs, stem orientation as removed.
5. Tighten the studs to the stated preload in three passes, cross pattern (SW-WO-5).
6. Remove blinds, restore the line-up, leak test the two joints on start-up (SW-TEST-2).

Affected items: 4"-P300-P-1021/A2, 4"-P300-P-1021/A1, 4"-P300-P-1021/A4

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1021	4"-P300-P-1021	4"-P300-P-1021	4	2 error(s), 0 warning(s) — see below

ISO-P-1021 (PDF) — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1021	1	error	sheet 1: CONT'D ON ISO-P-1021 SH 2 prints E 53'-6" N -13'-4 1/16" EL 109'-0" for E 53'-6" N -12'-7 15/16" EL 109'-0": the minus sign applies to the feet only; read as written the point is 8.12 in away (still on the centreline, at the wrong place along it)
ISO-P-1021	2	error	sheet 2: CONT'D FROM ISO-P-1021 SH 1 prints E 53'-6" N -13'-4 1/16" EL 109'-0" for E 53'-6" N -12'-7 15/16" EL 109'-0": the minus sign applies to the feet only; read as written the point is 8.12 in away (still on the centreline, at the wrong place along it)

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
V-102	equipment	V-102 vertical vessel 72" ID × 216" T/T		E 53'-6" N -25'-0" EL 100'-8-1/4"
4"-P300-P-1021/A2	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1021	E 53'-6" N -20'-8-3/16" EL 109'-0"
4"-P300-P-1021/A1	item	GASKET 4" P300 COMPRESSED SHEET	4"-P300-P-1021	E 53'-6" N -21'-3-7/8" EL 109'-0"
4"-P300-P-1021/A4	item	GASKET 4" P300 COMPRESSED SHEET	4"-P300-P-1021	E 53'-6" N -20'-1/2" EL 109'-0"
V-102-N2	nozzle	Nozzle V-102-N2 (outlet, 4" P300) on V-102		E 53'-6" N -21'-3-15/16" EL 109'-0"
4"-P300-P-1021/B2	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1021	E 44'-0" N 22'-6" EL 123'-10-13/16"
4"-P300-P-1021/B5	item	FLANGE WELD-NECK 4" P300 RAISED FACE	4"-P300-P-1021	E 44'-0" N 22'-6" EL 124'-9-5/16"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	equipment	V-102-N2	4"-P300-P-1021	Line end at nozzle V-102-N2: the equipment is out of service, drained and isolated under the shutdown isolation			
2	valve	4"-P300-P-1021/B2	4"-P300-P-1021	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1021/B2)	E 44'-0" N 22'-6" EL 123'-10-13/16"		
3	blind	4"-P300-P-1021/B5	4"-P300-P-1021	Fit a line blind between 4"-P300-P-1021/B2 and FLANGE WELD-NECK 4" P300 RAISED FACE (4"-P300-P-1021/B5), work side	E 44'-0" N 22'-6" EL 124'-6-9/16"		

Parts

Description	Qty	Replaces
VALVE, GATE, WEDGE, OS&Y 4" P300	1	4"-P300-P-1021/A2
GASKET 4" P300 COMPRESSED SHEET	1	4"-P300-P-1021/A1
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	1	4"-P300-P-1021/A3
GASKET 4" P300 COMPRESSED SHEET	1	4"-P300-P-1021/A4
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	1	4"-P300-P-1021/A6

Inspections

Id	Inspection	Criteria	Basis
WO-200-03-I1	Flange faces of both joints before the new gaskets go in	No radial scratch across the sealing face (SW-NDE-5)	SW-NDE-5

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
4"-P300-P-1021/A1	4 × 1" studs (4"-P300-P-1021/A3)	35,343	3	
4"-P300-P-1021/A4	4 × 1" studs (4"-P300-P-1021/A6)	35,343	3	

Test — Service leak test

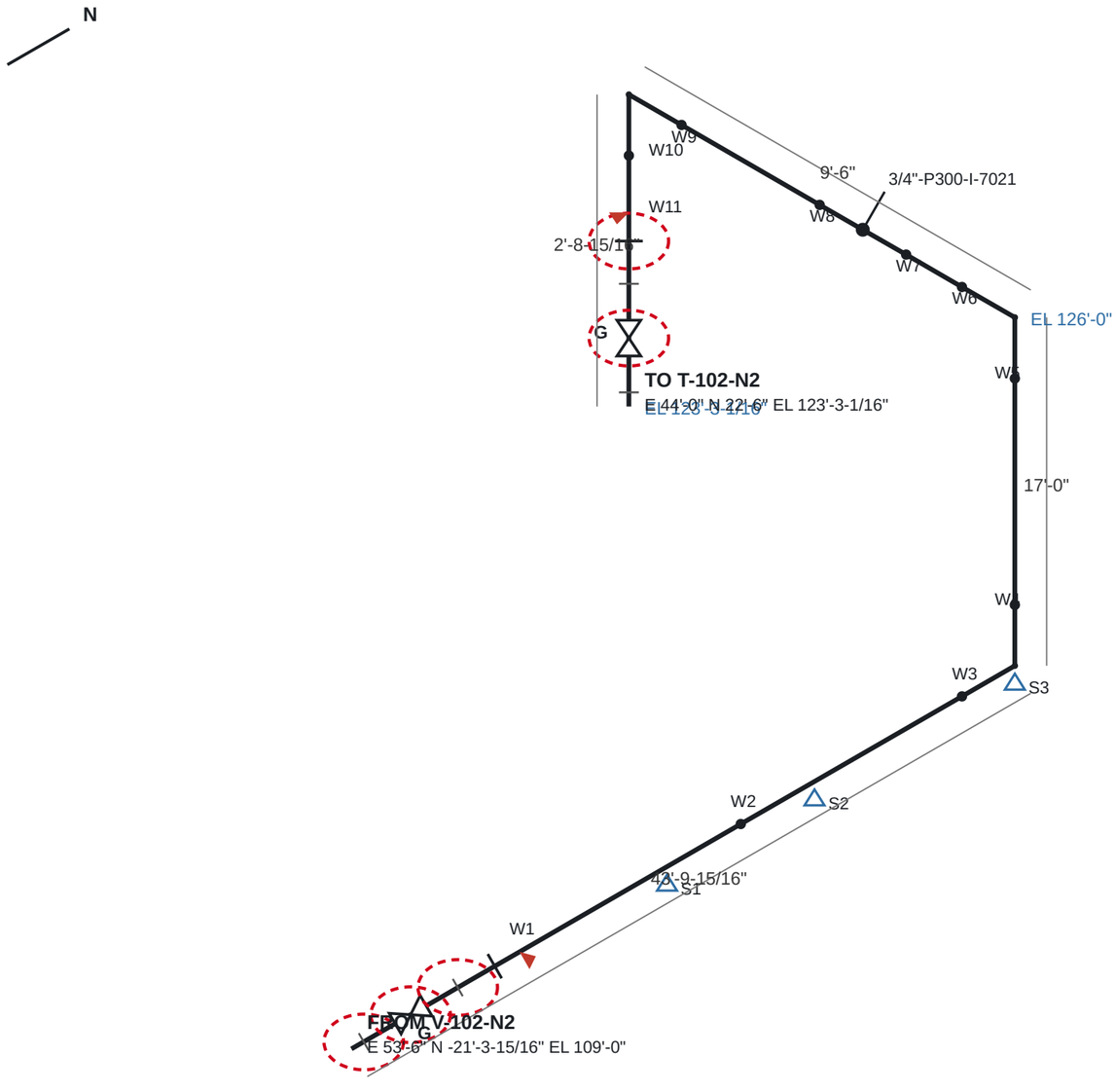
Description	Both remade joints checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Valve 4"-P300-P-1021/A2 removed; flange faces inspected	SW-NDE-5	No radial scratch across the sealing face	Flange face inspection record	W	W	—				
3	New valve verified: type, size, class, face-to-face and tag	GATE4-P300	Matches the parts list and the line class	Material receiving record	R	R	—				
4	New gaskets fitted, studs tightened to preload in three passes	SW-WO-5	Preload per stud as stated; three passes, cross pattern	Bolt-up record per joint	W	W	—				
5	Service leak test of the remade joints on start-up	SW-TEST-2	No leak at either joint	Leak test record	W	W	—				
6	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
7	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 4"-P300-P-1021 — Replace VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1021/A2)



BILL OF MATERIALS

1	GASKET 4" P300 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 4" P300	2
3	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	4
4	FLANGE WELD-NECK 4" P300 RAISED FACE	2
5	PIPE 4" P300 BEVELLED ENDS	65'-8-5/16"
6	ELBOW 90° R = 5/3 OD 4" P300	3
7	TEE REDUCING 4" × 3/4" P300	1

WELDS 11 (2 FIELD) · SUPPORTS 3 · SPOOLS 3

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11F

4"-P300-P-1021

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
V-102-N2 → T-102-N2
PIPE 65'-8-5/16" · WEIGHT 875 lb

DWG ISO-4"-P300-P-1021 REV 0 WO-200-03

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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